

$$KE_{\max} = \phi_1 - \phi_0$$

- 28 B** The highest energy level the hydrogen atom can reach is $n = 3$ if max KE is 12.5 eV.

The following transitions and the corresponding wavelengths are:

$$\lambda = hc/E = [(6.63 \times 10^{-34})(3.00 \times 10^8)/(1.60 \times 10^{-19})]/\Delta E \text{ in eV}$$

$$n = 3 \text{ to } n = 2: \lambda = 1.24 \times 10^{-6} / (3.40 - 1.53) = 6.65 \times 10^{-7} \text{ m (red)}$$

$$n = 3 \text{ to } n = 1: \lambda = 1.24 \times 10^{-6} / (13.6 - 1.53) = 1.03 \times 10^{-7} \text{ m}$$

$$n = 2 \text{ to } n = 1: \lambda = 1.24 \times 10^{-6} / (13.6 - 3.40) = 1.22 \times 10^{-7} \text{ m}$$

- 29 D** The positions of the peaks are characteristic of the energy levels in the target metal